

Strategy description:

- For every day, from 9:30am to 4pm (NY timezone) you are able to trade.
- Given a universe, in our case is Nasdaq100 stocks, calculate all the possible combination (pairs) by their rolling correlation (last 60 days) vs the other stocks in the same sector group. Once you have all the correlations per sector, you will make the top50 sorted by correlation, this top can include pairs from different sectors. You will define the leader as the stock in the pair with the most marketcap, and the follower is the other one. But note that the pairs per row in the top50 table should be the same sector as we calculate them like that before. It is impossible to have the correlation of two different stocks in different sectors in the top50 global ranking.
- Now, for every ticker in the top50 pairs universe, we compute the 15m sma. Note that as we want to trade as soon as we can (9:30-9:31) we must have computed the smas with data from 9:14-9:15.
- We will apply something that we are going to call an "edge factor" to every sma. Leader will have a 0.006 (0.6%) and follower will have a 0.002 (0.2%). So now, for the leader the new value of the SMA will be $SMA * (1 + \text{leader edge factor})$ (long sma). For the follower will be $SMA * (1 + \text{follower edge factor})$ (long sma).
- Once we have all this we will set the rules for buying. You will buy follower when the leader price is above their long SMA and the follower price is under their long SMA.
- You will close positions if the leader price drops below their base SMA (without any edge factor). or if the trading days is about to end, we close positions 5min before the trading day ends (at 3:55pm).
- For shorts: We will also apply edge factors to every sma. So now, for the leader the new value of the SMA will be $SMA * (1 - \text{leader edge factor})$ (short sma). For the follower will be $SMA * (1 - \text{follower edge factor})$ (short sma). Once we have all this we will set the rules for shorting. You will short follower when the leader price is below their short SMA and the follower price is under their short SMA. You will close positions if the leader price goes above their base SMA (without any edge factor). Or if the trading days is about to end, we close positions 5min before the trading day ends (at 3:55pm).
- You won't have any overnight position.
- Note that you only will be trading 50 different pairs of stocks.
- Use a fixed 100k notional per position.
- Trade only stocks with a price above \$10.